

Convergence Behavior of the Direct Extension to the ADMM

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Abstract. This report is based on the paper: The Direct Extension of ADMM for Multi-block Convex Minimization Problems is Not Necessarily Convergent [1].

The Alternating Direction Method of Multipliers (ADMM) has applications in many fields and has proven convergence results for two blocks of variables. The aforementioned paper analyzes the behavior of an extended ADMM model for multi-block convex minimization problems. In this report, we will examine the key results, examples, and consequences from this paper.

1 Introduction

The original ADMM model concerns solving convex minimization problems of the form:

$$\begin{aligned} & \min \theta_1(x_1) + \theta_2(x_2) \\ & \text{s.t. } A_1x_1 + A_2x_2 = b, \\ & x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2. \end{aligned} \tag{1}$$

The direct extension to the ADMM, aims to solve an extended problem, where the objective function is the sum of three functions without coupled variables:

$$\begin{aligned} & \min \theta_1(x_1) + \theta_2(x_2) + \theta_3(\mathbf{x}_3) \\ & \text{s.t. } A_1x_1 + A_2x_2 + \mathbf{A}_3\mathbf{x}_3 = \mathbf{b}, \\ & x_1 \in \mathcal{X}_1, x_2 \in \mathcal{X}_2, \mathbf{x}_3 \in \mathcal{X}_3. \end{aligned} \tag{2}$$

where:

- θ_i are closed convex functions
- $\mathcal{X}_i$ are closed convex sets
- and the solution set of (2) is assumed to be non-empty.

This extended model has many applications from the "image alignment problem" [2] to principal component analysis models with incomplete data [3]. Thus it is desirable to obtain results about its convergence properties.

Then as with the original ADMM, the Augmented Lagrangian is defined as:

$$\mathcal{L}_A(x_1, x_2, x_3, \lambda) = \sum_{i=1}^3 \theta_i(x_i) - \lambda^T \left(\sum_{i=1}^3 A_i x_i - b \right) + \frac{\beta}{2} \left\| \sum_{i=1}^3 A_i x_i - b \right\|^2 \tag{3}$$

The update scheme is as below, where each variable is solved in turn for each iteration k :

$$\begin{cases} x_1^{k+1} &= \arg \min \{ \mathcal{L}_A(x_1, x_2^k, x_3^k, \lambda^k) \mid x_1 \in X_1 \}, \\ x_2^{k+1} &= \arg \min \{ \mathcal{L}_A(x_1^{k+1}, x_2, x_3^k, \lambda^k) \mid x_2 \in X_2 \}, \\ x_3^{k+1} &= \arg \min \{ \mathcal{L}_A(x_1^{k+1}, x_2^{k+1}, x_3, \lambda^k) \mid x_3 \in X_3 \}, \\ \lambda^{k+1} &= \lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b). \end{cases} \quad (4)$$

2 Convergence

The paper shows that convergence of (4) is achieved when any two coefficient matrices are orthogonal. Further, this condition is shown to be sufficient without the requirement of any strong convexity assumptions.

The analysis will be split into two cases below:

2.1 Case 1: $A_1^T A_2 = 0$ or $A_2^T A_3 = 0$

This is when two coefficient matrices in consecutive order are orthogonal. For succinctness, we will look at $A_1^T A_2 = 0$, but the second setting is also similar.

By using the first order conditions for optimality, and substituting $A_1^T A_2 = 0$, we arrive at the following set of conditions:

$$\begin{cases} \theta_1(x_1) - \theta_1(x_1^{k+1}) + (x_1 - x_1^{k+1})^T \{-A_1^T [\lambda^k - \beta(A_1 x_1^{k+1} + A_3 x_3^k - b)]\} \geq 0, & \forall x_1 \in \mathcal{X}_1, \end{cases} \quad (5a)$$

$$\begin{cases} \theta_2(x_2) - \theta_2(x_2^{k+1}) + (x_2 - x_2^{k+1})^T \{-A_2^T [\lambda^k - \beta(A_2 x_2^{k+1} + A_3 x_3^k - b)]\} \geq 0, & \forall x_2 \in \mathcal{X}_2, \end{cases} \quad (5b)$$

$$\begin{cases} \theta_3(x_3) - \theta_3(x_3^{k+1}) + (x_3 - x_3^{k+1})^T \{-A_3^T [\lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b)]\} \geq 0, & \forall x_3 \in \mathcal{X}_3. \end{cases} \quad (5c)$$

However, it is noted that these are exactly the optimality conditions of the scheme:

$$\begin{cases} (x_1^{k+1}, x_2^{k+1}) = \arg \min \left\{ \theta_1(x_1) + \theta_2(x_2) - (\lambda^k)^T (A_1 x_1 + A_2 x_2) + \frac{\beta}{2} \|A_1 x_1 + A_2 x_2 + A_3 x_3^k - b\|^2 \mid (x_1, x_2) \in \mathcal{X}_1 \times \mathcal{X}_2 \right\}, & (6a) \\ x_3^{k+1} = \arg \min \left\{ \theta_3(x_3) - (\lambda^k)^T A_3 x_3 + \frac{\beta}{2} \|A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3 - b\|^2 \mid x_3 \in \mathcal{X}_3 \right\}, & (6b) \\ \lambda^{k+1} = \lambda^k - \beta(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b). & (6c) \end{cases}$$

Thus, this is a specific application of the original ADMM, if we regard (x_1, x_2) as one variable, $[A_1, A_2]$ as one matrix, and $\theta_1(x_1) + \theta_2(x_2)$ as one function.

So the existing convergence results hold (which we discuss later in this section).

2.2 Case 2: $A_1^T A_3 = 0$

In order to prove convergence of this case, we first have to establish some propositions and notational changes.

Firstly, we note that the update order in (4) is cyclical, $x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow \lambda$ so equivalently, we can update via: $x_2 \rightarrow x_3 \rightarrow \lambda \rightarrow x_1$:

$$\begin{cases} x_2^{k+1} &= \arg \min \{ \mathcal{L}_A(x_1^k, x_2, x_3^k, \lambda^k) \mid x_2 \in \mathcal{X}_2 \}, \\ x_3^{k+1} &= \arg \min \{ \mathcal{L}_A(x_1^k, x_2^{k+1}, x_3, \lambda^k) \mid x_3 \in \mathcal{X}_3 \}, \\ \lambda^{k+1} &= \lambda^k - \beta (A_1 x_1^k + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b), \\ x_1^{k+1} &= \arg \min \{ \mathcal{L}_A(x_1, x_2^{k+1}, x_3^{k+1}, \lambda^{k+1}) \mid x_1 \in \mathcal{X}_1 \}. \end{cases} \quad (7)$$

And as with Case 1, by considering the first order optimality conditions, and substituting $A_1^T A_3 = 0$, we get the following:

$$\begin{cases} \theta_2(x_2) - \theta_2(x_2^{k+1}) + (x_2 - x_2^{k+1})^T \left\{ -A_2^T \lambda^{k+1} + \beta A_2^T A_3 (x_3^k - x_3^{k+1}) \right\} \geq 0, \\ \theta_3(x_3) - \theta_3(x_3^{k+1}) + (x_3 - x_3^{k+1})^T \left\{ -A_3^T \lambda^{k+1} \right\} \geq 0, \\ \left(A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1} - b \right) + A_1 (x_1^k - x_1^{k+1}) - \frac{1}{\beta} (\lambda^k - \lambda^{k+1}) = 0, \\ \theta_1(x_1) - \theta_1(x_1^{k+1}) + (x_1 - x_1^{k+1})^T \left\{ -A_1^T \lambda^{k+1} - \beta A_1^T A_1 (x_1^k - x_1^{k+1}) + A_1^T (\lambda^k - \lambda^{k+1}) \right\} \geq 0. \end{cases} \quad (8)$$

Remark 1. The extended ADMM model can be characterized by a variational inequality, in the sense that finding a saddle point of the Lagrangian (3) is equivalent to solving the following variational problem:

Find $w^* \in \Omega$ such that

$$\text{VI}(\Omega, F, \theta) : \theta(u) - \theta(u^*) + (w - w^*)^T F(w^*) \geq 0, \quad \forall w \in \Omega, \quad (9)$$

where

$$u = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad w = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda \end{pmatrix}, \quad \theta(u) = \theta_1(x_1) + \theta_2(x_2) + \theta_3(x_3),$$

and

$$F(w) = \begin{pmatrix} -A_1^T \lambda \\ -A_2^T \lambda \\ -A_3^T \lambda \\ A_1 x_1 + A_2 x_2 + A_3 x_3 - b \end{pmatrix}.$$

It should be noted that the mapping F is monotone as it is affine and skew-symmetric.

Next, the paper establishes two lemmas to rewrite (8) in a compact form using the remark above and assuming $A_1^T A_3 = 0$. These lead to the following lemma:

Lemma 1.

Let $w^{k+1} = (x_1^{k+1}, x_2^{k+1}, x_3^{k+1}, \lambda^{k+1})$ be generated by (7) from given $v^k = (x_1^k, x_3^k, \lambda^k)$.

If $A_1^T A_3 = 0$, we have $\tilde{w}^k \in \Omega$ and

$$\theta(u) - \theta(\tilde{u}^k) + (w - \tilde{w}^k)^T F(\tilde{w}^k) \geq \frac{1}{2} (\|v - v^{k+1}\|_H^2 - \|v - v^k\|_H^2) + \frac{1}{2} \|v^k - v^{k+1}\|_H^2, \quad \forall w \in \Omega,$$

where $\tilde{w}^k$ and $\tilde{u}^k$ are the auxiliary sequences defined below:

$$\tilde{w}^k := \begin{pmatrix} \tilde{x}_1^k \\ \tilde{x}_2^k \\ \tilde{x}_3^k \\ \tilde{\lambda}^k \end{pmatrix} := \begin{pmatrix} x_1^{k+1} \\ x_2^{k+1} \\ x_3^{k+1} \\ \lambda^{k+1} - \beta A_3(x_3^k - x_3^{k+1}) \end{pmatrix}, \quad \text{and} \quad \tilde{u}^k := \begin{pmatrix} \tilde{x}_1^k \\ \tilde{x}_2^k \\ \tilde{x}_3^k \end{pmatrix},$$

Note: the norm in the inequality is with respect to matrix H which is defined as:

$$H = \begin{pmatrix} \beta A_1^T A_1 & 0 & -A_1^T \\ 0 & \beta A_3^T A_3 & 0 \\ -A_1 & 0 & \frac{1}{\beta} I \end{pmatrix}$$

From here, we can establish the theorem about convergence for case 2:

Theorem 1. Assume $A_1^T A_3 = 0$ for the extended ADMM model. Let $\{x_1^k, x_2^k, x_3^k, \lambda^k\}$ be the sequence generated by the direct extension of ADMM (4). Then, we have:

- (i) The sequence $\{v^k := (x_1^k, x_3^k, \lambda^k)\}$ is contractive with respect to the solution of $VI(\Omega, F, \theta)$, i.e.,

$$\|v^{k+1} - v^*\|_H^2 \leq \|v^k - v^*\|_H^2 - \|v^k - v^{k+1}\|_H^2.$$

- (ii) If the matrices $[A_1, A_2]$ and A_3 are assumed to be full column rank, then the sequence $\{w^k\}$ converges to a KKT point of the model (4).

The proof of the theorem is detailed in the paper, however we outline a sketch and key ideas below.

Sketch of Proof.

- (i) From the Lemma, we have that

$$\theta(u) - \theta(\tilde{u}^k) + (w - \tilde{w}^k)^T F(\tilde{w}^k) \geq \frac{1}{2} (\|v - v^{k+1}\|_H^2 - \|v - v^k\|_H^2) + \frac{1}{2} \|v^k - v^{k+1}\|_H^2, \quad \forall w \in \Omega,$$

By setting $w = w^*$ (and thus $v = v^*$, $u = u^*$), and rearranging we have:

$$\begin{aligned} \frac{1}{2} (\|v^k - v^*\|_H^2 - \|v^{k+1} - v^*\|_H^2) - \frac{1}{2} \|v^k - v^{k+1}\|_H^2 &\geq \theta(\tilde{u}^k) - \theta(u^*) + (\tilde{w}^k - w^*)^T F(\tilde{w}^k) \\ &\geq \theta(\tilde{u}^k) - \theta(u^*) + (\tilde{w}^k - w^*)^T F(w^*) \\ &\geq 0 \end{aligned}$$

Where the second inequality came from the monotonicity of F , and the third inequality comes from the variational inequality ($VI(\Omega, F, \Theta)$) established in (9)). Now rearranging, we get the following inequality:

$$\|v^k - v^*\|_H^2 \leq \|v^{k+1} - v^*\|_H^2 + \|v^k - v^{k+1}\|_H^2$$

This shows that v^k is contractive with respect to the solution set of $VI(\Omega, F, \Theta)$.

(ii) Key Ideas

Using the contractive inequality from (i), it is shown that the sequences $\{A_1 x_1^k - \frac{1}{\beta} \lambda^k\}$ and $\{A_3 x_3^k\}$ are bounded.

Note that: $A_1 x_1^k + A_2 x_2^k = (A_1 x_1^k - \frac{1}{\beta} \lambda^k) - (A_1 x_1^{k-1} - \frac{1}{\beta} \lambda^{k-1}) - A_3 x_3^k + b$, implies that $\{A_1 x_1^k + A_2 x_2^k\}$ is bounded.

The assumption that $[A_1, A_2, A_3]$ have full column rank then implies that $\{x_1^k\}, \{x_2^k\}, \{x_3^k\}, \{\lambda^k\}$ are bounded.

Thus, there exists a subsequence $\{x_1^{n_k+1}, x_2^{n_k+1}, x_3^{n_k+1}, \lambda^{n_k+1}\}$ that converges to a limit point: $(x_1^\infty, x_2^\infty, x_3^\infty, \lambda^\infty)$.

By further taking limits, it can be shown that $(x_1^\infty, x_2^\infty, x_3^\infty, \lambda^\infty)$ satisfies the $VI(\Omega, F, \Theta)$, and thus is a KKT point of the model.

Finally, it is shown that:

$$\lim_{k \rightarrow \infty} x_1^k = x_1^\infty, \quad \lim_{k \rightarrow \infty} x_2^k = x_2^\infty, \quad \lim_{k \rightarrow \infty} x_3^k = x_3^\infty, \quad \lim_{k \rightarrow \infty} \lambda^k = \lambda^\infty,$$

which establishes the theorem. $\square$

2.3 Convergence Rates

So now we have seen that when specific orthogonality conditions on the coefficient matrices are imposed, convergence can be achieved. The paper goes on to establish a worst case convergence rate for the direct extension of the ADMM, which is established in the theorem below:

Theorem 2. Assume $A_1^T A_3 = 0$ for the model (2). Let $\{(x_1^k, x_2^k, x_3^k, \lambda^k)\}$ be the sequence generated by the direct extension of ADMM (7) and $\tilde{w}^k$ be defined as above. After t iterations, we take

$$\tilde{w}_t = \frac{1}{t+1} \sum_{k=0}^t \tilde{w}^k.$$

Then, $\tilde{w} \in W$ and it satisfies

$$\theta(\tilde{w}_t) - \theta(w) + (\tilde{w}_t - w)^T F(w) \leq \frac{1}{2(t+1)} (\|v - v^0\|_H^2), \quad \forall w \in \Omega.$$

So now we can compare the rates of convergence of different ADMM models:

- Traditional ADMM has a convergence rate of $O(1/k)$ [4]
- Modified ADMM, with f and g strongly convex, and g quadratic has a convergence rate of $O(1/k^2)$ [5]
- Direct Extension to the ADMM has an ergodic convergence rate of $O(1/k)$ that is, after k iterations, $\frac{1}{k} \sum_{i=1}^k (x_1^k, x_2^k, x_3^k, \lambda^k) \rightarrow (x_1^*, x_2^*, x_3^*, \lambda^*)$ [1]

3 Divergence

3.1 Constructing a Divergent Example

Although we have shown that there exist sufficient conditions for convergence of extended ADMM, the orthogonality conditions are not always practical in practice. So we will construct an example where divergence occurs when such orthogonality conditions are missing.

We examine the following simple linear homogeneous equation with three variables:

$$A_1 x_1 + A_2 x_2 + A_3 x_3 = 0 \quad (10)$$

with the assumptions that $A_i \in \mathbb{R}^3$ ($i = 1, 2, 3$) are column vectors, the matrix $[A_1, A_2, A_3]$ is non singular, and each x_i is $\in \mathbb{R}$.

It is clear to see that the unique solution to (10) is $x_1 = x_2 = x_3 = 0$. Further, we can regard (10) as a specific version of the extended problem (3) with a null objective. That is:

$$\begin{aligned} \min & 0(x_1) + 0(x_2) + 0(x_3) \\ \text{s.t.} & A_1 x_1 + A_2 x_2 + A_3 x_3 = 0 \end{aligned} \quad (11)$$

Applying the first order optimality conditions to the Lagrange multipliers, and introducing a new variable $\mu^k := \lambda^k / \beta$, we have:

$$\begin{cases} -A_1^T \mu^k + A_1^T (A_1 x_1^{k+1} + A_2 x_2^k + A_3 x_3^k) = 0, & (12a) \\ -A_2^T \mu^k + A_2^T (A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^k) = 0, & (12b) \\ -A_3^T \mu^k + A_3^T (A_1 x_1^{k+1} + A_2 x_2^k + A_3 x_3^{k+1}) = 0, & (12c) \\ (A_1 x_1^{k+1} + A_2 x_2^{k+1} + A_3 x_3^{k+1}) + \mu^{k+1} - \mu^k = 0. & (12d) \end{cases}$$

Re-arranging (12a) we find:

$$x_1^{k+1} = \frac{1}{A_1^T A_1} (-A_1^T A_2 x_2^k - A_1^T A_3 x_3^k + A_1^T \mu^k) \quad (13)$$

Now, substituting (13) into (12b-d), we arrive at the following system of equations:

$$\begin{aligned}
& \underbrace{\begin{pmatrix} A_2^T A_2 & 0 & 0_{1 \times 3} \\ A_3^T A_2 & A_3^T A_3 & 0_{1 \times 3} \\ A_2 & A_3 & I_{3 \times 3} \end{pmatrix}}_L \begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} \\
&= \underbrace{\left[\begin{pmatrix} 0 & -A_2^T A_3 & A_2^T \\ 0 & 0 & A_3^T \\ 0_{3 \times 1} & 0_{3 \times 1} & I_{3 \times 3} \end{pmatrix} - \frac{1}{A_1^T A_1} \begin{pmatrix} A_2^T A_1 \\ A_3^T A_1 \\ A_1 \end{pmatrix} (-A_1^T A_2, -A_1^T A_3, A_1^T) \right]}_R \begin{pmatrix} x_2^k \\ x_3^k \\ \mu^k \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} = \underbrace{L^{-1} R}_M \begin{pmatrix} x_2^k \\ x_3^k \\ \mu^k \end{pmatrix}
\end{aligned}$$

So now the iterative scheme is:

$$\begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} = M \begin{pmatrix} x_2^k \\ x_3^k \\ \mu^k \end{pmatrix} = \dots = M^{k+1} \begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu^0 \end{pmatrix} \quad (14)$$

It should be noted that the matrix M defined above is independent of $\beta > 0$, so if the scheme is convergent for a selected $\beta > 0$ it is convergent for all $\beta > 0$, and similarly if it is divergent for a single $\beta > 0$ it is divergent for all $\beta > 0$. So we set $\beta = 1$ without loss of generality.

We note that convergence is achieved if the matrix mapping is a contraction, which is true if and only if the spectral radius of M , $\rho(M)$, is strictly less than 1.

3.2 A Specific Example

Now that we have an idea of a scenario which leads to divergence, we can create a specific example to show divergence. In particular we consider a matrix A such that $\rho(M) > 1$, and that any two coefficient matrices are non-orthogonal:

$$\text{Let } A = (A_1, A_2, A_3) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix}$$

Then, computing the matrix M from before, we get:

$$M = L^{-1} R = \frac{1}{162} \begin{pmatrix} 144 & -9 & -9 & -9 & 18 \\ 8 & 157 & -5 & 13 & -8 \\ 64 & 122 & 122 & -58 & -64 \\ 56 & -35 & -35 & 91 & -56 \\ -88 & -26 & -26 & -62 & 88 \end{pmatrix}$$

Which admits the eigenvalue decomposition:

$$M = V \text{Diag}(d) V^{-1}, \quad \text{where } d = \begin{pmatrix} 0.9836 + 0.2984i \\ 0.9836 - 0.2984i \\ 0.8744 + 0.2310i \\ 0.8744 - 0.2310i \\ 0 \end{pmatrix}$$

From here, we note that $\rho(M) = |d_1| = |d_2| > 1$.

It remains to be determined whether there are any real-valued non-convergent starting points.

Indeed, define the initial point $(x_2^0, x_3^0, \mu^0) := V(:, 1) + V(:, 2) \in \mathbb{R}^5$, then:

$$\begin{aligned} \begin{pmatrix} x_2^{k+1} \\ x_3^{k+1} \\ \mu^{k+1} \end{pmatrix} &= [V \text{Diag}(d) V^{-1}]^{k+1} \begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu^0 \end{pmatrix} = V \text{Diag}(d^{k+1}) V^{-1} \begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu^0 \end{pmatrix} \\ &= V \text{Diag}(d^{k+1}) \begin{pmatrix} 1 \\ 1 \\ 0_{3 \times 1} \end{pmatrix} \\ &= V \begin{pmatrix} (0.9836 + 0.2984i)^{k+1} \\ (0.9836 - 0.2984i)^{k+1} \\ 0_{3 \times 1} \end{pmatrix} \end{aligned}$$

which is divergent.

3.3 Main Result

The previous discussion leads us to the main theorem of the paper:

Theorem 3. *For the three-block convex minimization problem (2), there is an example where the direct extension of ADMM (4) is divergent for any penalty parameter $\beta > 0$ and for any starting point in a continuously dense half space of dimension 3.*

In our example, setting the initial point to be: $\begin{pmatrix} x_2^0 \\ x_3^0 \\ \mu_1^0 \\ \mu_2^0 \\ \mu_3^0 \end{pmatrix} = V \begin{pmatrix} \alpha_1 \\ \alpha_1 \\ \alpha_2 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$, the points with

$(\alpha_1 > 0, \alpha_2, \alpha_3) \in \mathbb{R}^3$ form a continuously dense half space of non-convergent starting points.

4 Extensions

4.1 Strong Convexity

Remark 2. If all functions θ_i in (4) are assumed to be strongly convex and the penalty parameter β is restricted to a specific range determined by the strong convex modulus of these functions, then the direct extension of ADMM is convergent. [6]

This naturally leads us an interesting question, is the extended ADMM scheme still convergent for a strongly convex optimization model when the restriction on β is removed?

The answer is no, which is evidenced by the following strongly convex minimization problem:

$$\begin{aligned} & \min 0.05x_1^2 + 0.05x_2^2 + 0.05x_3^2 \\ & \text{s.t.} \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \end{aligned} \quad (15)$$

In a similar calculation to before, the spectral radius for the relevant mapping matrix for (15) with $\beta = 1$ is $\rho(M) = 1.0087 > 1$. Thus again one can find a starting point such that the scheme diverges. This leads us to the next theorem:

Theorem 4. *For the model (2) with the strong convex assumption on its objective function, the direct extension of ADMM (4) is not necessarily convergent for all $\beta > 0$.*

4.2 Small Step-Size Variant

Adaptations and extensions of existing models in order to obtain desired properties such as stronger or faster convergence are common practices in optimization. One such example is the ADMM Variant with a Small Step-Size, which was proposed [7] to deal with convergence ambiguity by attaching a shrinkage step-size factor to the Lagrange-multiplier update:

$$\lambda^{k+1} = \lambda^k - \gamma\beta \left(\sum_{i=1}^3 A_i x_i^k - b \right) \quad (16)$$

It has been shown that when γ is sufficiently small, and certain regularity assumptions are satisfied, then the ADMM variant is linearly convergent.[7] However, the requirement on γ depends on both the objective function and the problem data.

Again, this leads us to a question, does a "problem-data-independent" interval of γ exist such that this ADMM variant is convergent?

Whilst a theoretical answer does not exist, the paper conjectures (through the use of numerical experiments) that the answer is no.

This was concluded by considering the same linear equation (10), but now using the matrix A below with small positive scalar $\gamma > 0$:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 + \gamma \\ 1 & 1 + \gamma & 1 + \gamma \end{pmatrix}.$$

Denoting $M(\gamma)$ as the mapping matrix for this ADMM variant, it is numerically computed that from a certain initial point, $\rho(M(\gamma)) > 1$ with values of γ ranging from 1 to 1×10^{-8} (see Table 1). That is, this scheme remains divergent even if the step size factor γ is very small.

Table 1. Spectral radius of the small step-size variant of ADMM ($\beta = 1$)

γ	1	0.1	1e-2	1e-3	1e-4	1e-5	1e-6	1e-7	1e-8
$\rho(M(\gamma))$	1.027839	1.002637	1.000105	1.000004	> 1	> 1	> 1	> 1	> 1

5 Conclusions

In this report we have seen how the traditional alternating direction method of multiplier (ADMM) can have a “direct extension” using three blocks of variables and functions. We have examined how orthogonality conditions on the coefficient matrices are sufficient to ensure provable convergence. By construction of an example, we have also observed that the direct extension is not necessarily convergent in general. Additionally, in extensions to existing publications, this paper shows that even strong convexity assumptions do not guarantee convergence for the direct extension.

Overall the findings in this paper serve to fill in gaps in the literature regarding open questions about the ADMM. Further, the results justify the correction steps and adjustments to algorithms in recent work [8] to produce an algorithm with more desirable convergence results.

References

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